

Assignment No: 2(A)

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#include <iostream>
#include <omp.h>

using namespace std;

void sequentialBubbleSort(int arr[], int n)
{
    for(int i=0;i<n-1;i++)
    {
        for(int j=0;j<n-i-1;j++)
        {
            if(arr[j] > arr[j+1])
            {
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

void parallelBubbleSort(int arr[], int n)
{
    for(int i=0;i<n;i++)
    {
        // Even phase
        #pragma omp parallel for
        for(int j=0;j<n-1;j+=2)
        {
            if(arr[j] > arr[j+1])
            {
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }

        // Odd phase
        #pragma omp parallel for
        for(int j=1;j<n-1;j+=2)
        {
            if(arr[j] > arr[j+1])
            {
                int temp = arr[j];
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        arr[j] = arr[j+1];
        arr[j+1] = temp;
    }
}

}

void printArray(int arr[], int n)
{
    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
    cout<<endl;
}

int main()
{
    int n;

    cout<<"Enter number of elements: ";
    cin>>n;

    int arr[n], arr2[n];

    cout<<"Enter elements:\n";

    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
        arr2[i] = arr[i]; // copy for parallel
    }

    double start, end;

    // Sequential timing
    start = omp_get_wtime();
    sequentialBubbleSort(arr,n);
    end = omp_get_wtime();

    cout<<"\nSequential Sorted Array:\n";
    printArray(arr,n);
    cout<<"Sequential Time: "<<end-start<<" seconds\n";

    // Parallel timing
    start = omp_get_wtime();
    parallelBubbleSort(arr2,n);
    end = omp_get_wtime();

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    cout<<"\nParallel Sorted Array:\n";  
    printArray(arr2,n);  
    cout<<"Parallel Time: "<<end-start<<" seconds\n";  
  
    return 0;  
}
```

Input:

Enter number of elements: 6

Enter elements:

9 3 7 1 5 2

Output:

Sequential Sorted Array:

1 2 3 5 7 9

Sequential Time: 0.00002 seconds

Parallel Sorted Array:

1 2 3 5 7 9

Parallel Time: 0.00001 seconds